

# Soohyeon Choi

Research Scientist | AI and LLM Safety and Security

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## Profile

AI and LLM security researcher with a Ph.D. in Computer Science and more than five years of research experience. Investigates how advanced language models fail under adversarial use and how generated code can be detected, attributed, and provenance-marked. Research connects compositional jailbreaking, exposure-aware evaluation, human and AI-agent code attribution, and post-hoc watermarking around a common goal of reliable measurement under adversarial and deployment shifts. Builds reproducible, large-scale benchmarks and evaluation pipelines using fine-tuned code encoders, zero- and few-shot LLMs, matched controls, robustness tests, and statistical auditing across Python, Java, and C++. First-author publications include IEEE TDSC, IEEE ICDCS, ICICS, and CSoNet. Research has received 295 Google Scholar citations as of August 2026.

## Research Experience

### Singapore Management University

Research Scientist I

Singapore

May 2025–Present

- Conduct research on AI and LLM security, including compositional jailbreak attacks, exposure-aware generated-code detection, AI coding-agent attribution, and post-hoc code watermarking.
- Design large-scale evaluation pipelines using fine-tuned code encoders, zero-shot detectors, robustness tests, matched controls, and statistically audited datasets.
- Collaborate with academic and industry researchers across Singapore and the United States, and communicate findings through manuscripts, artifacts, and research presentations.

### University of Central Florida

Graduate Research Assistant, SEAL Lab

Orlando, Florida, USA

Aug. 2021–May 2025

- Developed machine-learning methods to identify ChatGPT-generated and ChatGPT-transformed source code and evaluated their robustness to code transformation and distribution shift.
- Built a sequence-to-sequence code-transformation approach for authorship evasion and studied adversarial robustness in language and sensor-based classification systems.
- Published first-authored work in IEEE TDSC, IEEE ICDCS, ICICS, and CSoNet while completing a Ph.D. focused on machine learning, LLMs, and security.

### A\*STAR Institute for Infocomm Research

Research Officer

Singapore

Dec. 2023–Dec. 2024

- Developed zero-shot, few-shot, and tournament-style LLM methods for source-code authorship attribution, evaluating cross-language generalization and adversarial robustness across C++ and Java.

### South Dakota State University

Graduate Research Assistant, CCT Lab

Brookings, South Dakota, USA

Aug. 2019–Dec. 2020

- Developed a low-computation authentication protocol for wireless body-area networks using elliptic-curve cryptography and two-dimensional hash chains.

## Education

**Ph.D. in Computer Science**, University of Central Florida

2021–2025

Advisor: Prof. David Mohaisen. Focus: machine learning, LLMs, and security.

Orlando, USA

**M.S. in Computer Science**, South Dakota State University

2018–2021

Advisor: Prof. Sung Shin. Focus: authentication and sensor-network security.

Brookings, USA

**B.E. in Computer Engineering**, Keimyung University

2011–2017

Daegu, South Korea

## Selected AI Safety and Security Research

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### Compositional Jailbreaking of Aligned LLMs

Corresponding author; under review, 2026

- Evaluated all ordered pairs of 12 jailbreak mutators against three LLMs to identify synergistic, destructive, and structurally incompatible attack compositions; developed completeness and validity measures for attack persistence and effectiveness.

### CommitTrace: AI Coding-Agent Attribution

First author; under review, 2026

- Diagnosed why prior detectors fail on repository-local changes, with zero-shot AUROC ranging from 0.398 to 0.528, then developed a code-only representation combining a fine-tuned encoder, character n-grams, and stylometry.
- Achieved binary macro F1 of 0.966 on file patches and 0.953 on hunks across five coding agents; evaluated fragmentation, context, robustness, cross-language transfer, and repository-disjoint generalization.

### Talking to Themselves: Exposure-Aware Code Detection

First author; under review, 2026

- Evaluated heuristic and LLM-based generated-code detectors across C++, Java, and Python; found heuristic AUROC falling from 0.97 on fresh GitHub data to 0.15 on exposed CodeNet data and GPT-4.1 accuracy falling from 93% to 2% after copyright-header insertion.

### I Can Find You in Seconds: LLM Code Authorship Attribution

First author; under review, 2026

- Developed zero-shot, few-shot, and tournament-style LLM methods for code authorship attribution, achieving MCC up to 0.78 and scaling to 500 C++ and 686 Java authors with one reference sample per author.

### Multi-Channel Spread-Spectrum Code Watermarking

First author; under review, 2026

- Developed a post-hoc, training-free method that embeds a 24-bit provider label through naming and structural channels with explicit erasure-oriented recovery.
- Evaluated 1,750 Python files with 100% clean target-label recovery and 94.1% recovery under 10% random per-site corruption in the tested setting; embedding and detection run on CPU without access to the generating model.

## Selected Peer-Reviewed Publications

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- **S. Choi** and D. Mohaisen, “Attributing ChatGPT-Generated Source Codes,” *IEEE Transactions on Dependable and Secure Computing*, 2025.
- **S. Choi** et al., “Attributing ChatGPT-Transformed Synthetic Code,” *IEEE International Conference on Distributed Computing Systems*, 2025.
- M. F. Kharma, **S. Choi**, M. Alkhanafseh, and D. Mohaisen, “Security and Quality in LLM-Generated Code: A Multi-Language, Multi-Model Analysis,” *IEEE Transactions on Dependable and Secure Computing*, 2026.
- A. Alkinoon, T. C. Dang, A. Alghuried, A. Alghamdi, **S. Choi**, et al., “A Comprehensive Analysis of Evolving Permission Usage in Android Apps: Trends, Threats, and Ecosystem Insights,” *Journal of Cybersecurity and Privacy*, 2025.
- **S. Choi**, R. Jang, D. Nyang, and D. Mohaisen, “Untargeted Code Authorship Evasion with Seq2Seq Transformation,” *CSoNet*, 2023.
- **S. Choi**, M. Mohaisen, D. Nyang, and D. Mohaisen, “Revisiting the Deep Learning-Based Eavesdropping Attacks via Facial Dynamics from VR Motion Sensors,” *ICICS*, 2023.
- M. Omar, **S. Choi**, D. Nyang, and D. Mohaisen, “Quantifying the Performance of Adversarial Training on Language Models with Distribution Shifts,” *CySSS*, 2022.
- M. Omar, **S. Choi**, D. Nyang, and D. Mohaisen, “Robust Natural Language Processing: Recent Advances, Challenges, and Future Directions,” *IEEE Access*, 2022.

## Technical Skills and Professional Service

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**Programming and tools:** Python, C/C++, Java, SQL, Linux, Git, L<sup>A</sup>T<sub>E</sub>X.

**Machine learning:** PyTorch, TensorFlow, scikit-learn, Transformers, model fine-tuning, LLM evaluation, adversarial ML, benchmark design, statistical evaluation.

**Reviewing:** IEEE TIFS (2025–2026), IEEE TDSC (2023–2026), ACM AsiaCCS (2025), IEEE TMC (2022), and *Computer Networks* (2022).